Using Network Statistics to Analyze Amazon Movie Reviews

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Abstract

Online reviews can significantly alter consumer behavior, especially on platforms like Amazon where user-generated content helps shape product perception. While traditional approaches focus on sentiment analysis or topic modeling, they often overlook the relational structure between the products and reviewers. In our study, we apply network analysis to uncover deeper patterns in Amazon movie reviews using a dataset obtained from the Stanford Network Analysis Project, which spans over a decade and contains millions of reviews. The goal is to uncover how reviewers interact with products and with each other, and identify meaningful relationships or patterns in movie reviews data. We construct a bipartite graph of users and movies, which is then projected into a movie network where nodes (movies) are connected if reviewed by the same user. We then explore the network's characteristics through EDA and Exponential Random Graph Models (ERGMs). Our analysis reveals that the network is very compact, suggesting that many movies have been reviewed by the same Amazon user. We hope that this research can contribute to methodological approaches for analyzing large-scale bipartite networks and provides a foundation for further research in recommender systems, marketing, and behavioral analysis.

1 Introduction

Online reviews play an important role in shaping consumer behavior on platforms such as Amazon. These reviews not only shape product perceptions and influence purchasing decisions, but also provide valuable insights into market trends and consumer sentiment. While traditional approaches like sentiment analysis and topic modeling are commonly used to analyze reviews, they often overlook the underlying relational structure between products and reviewers.

Understanding this structure is crucial, as it reveals how products are connected through shared audiences and how user behavior reflects broader patterns in consumption. Prior research has shown the benefits of incorporating network perspectives: graph-based anomaly detection has been used to uncover fraudulent review behavior (Tang et al., 2013), and models that integrated sentiment analysis have improved recommendation performance by capturing user preferences (Osman, 2016). Some studies have gone further by constructing product-word bipartite graphs to group products with similar review language, or by building product-reviewer networks to improve recommendation systems (Crawford & Cho, 2015).

Building on this foundation, we examine the structure of Amazon movie reviews through network analysis. Using the Amazon movie review dataset from the Stanford Network Analysis Project (Leskovec and Krevl, 2014), we construct a user–movie bipartite graph and project it into a movie–movie network, where connections represent shared reviewers. This projection enables us to study the network's structure through exploratory data analysis and Exponential Random Graph Models (ERGMs). The resulting network is relatively compact, with many movies connected by overlapping reviewer communities. By applying these methods, we aim to better understand patterns in user behavior and provide practical insights for recommender systems, marketing strategies, and consumer research.

2 Network Construction

2.1 Network Fundamentals

The Amazon movie review data is naturally represented as a bipartite network. In such networks, there are two distinct types of nodes, and edges exist only between nodes of different types, never within the same type (Luke, 2015). In this case, the two node types are users and movies, with an edge connecting a user to a movie if the user has reviewed it.

2.2 Bipartite Projection

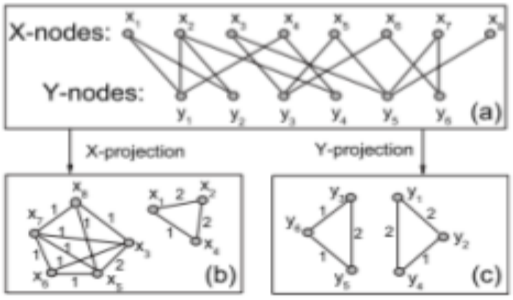
To analyze relationships between entities of the same type, bipartite projection is employed. This technique transforms a bipartite network into a one-mode network, where nodes of the same type are connected if they share a common neighbor in the other set (Luke, 2015).

In our case, we project onto the movie set to create a movie–movie network, where two movies are connected if reviewed by at least one common user. The resulting network is undirected and weighted, with edge weights representing the number of users who have reviewed both movies.

For example, consider Figure 1, which illustrates the original bipartite network, where X-nodes represent users and Y-nodes represent movies. *Movie y1* and *Movie y2* have two reviewers, *x1* and *x2*, in common, so the edge between them has a weight of 2. *Movie y2* and *Movie y4* share only one reviewer, so their edge has a weight of 1. If two movies do not share any reviewers, such as *Movie y2* and *Movie y5,* no edge is formed between them.

Figure 1

Illustration of a Bipartite Projection



Note. Zhou et al. (2007), "Bipartite Network Projection and Personal Recommendation," Physical Review E, 76(4): 046115. https://doi.org/10.1103/PhysRevE.76.046115

We also generated a user–user projection, where users are connected based on shared movie reviews, and edge weights reflect the number of movies reviewed in common. However, this network was comparatively sparse. In contrast, the movie projection formed a denser and more cohesive network with clearer community structure. Given its richer connectivity, we focused our subsequent analysis on the movie–movie network, which more effectively reflects patterns of shared audience engagement and potential viewing similarity.

3 Data

3.1 Raw Dataset

The dataset contains all movie reviews posted on Amazon from August of 1997 until October 2012. This amounts to 7,911,684 reviews, written by 889,176 users about 253,059 movies. Out of all of the reviewers, 16,341 wrote 50 or more reviews.

The chosen dataset has the variables described in Table 1:

Table 1

Name and description of each one of the variables in the original dataset

Variable Name	Explanation	Sample Value
product/productId	This is a unique identifying number for the movie	B000ZLFALI
review/userId	This is a unique identifying number for the user.	A1AWYPH2UEHNYH
review/profileName	Name/Username of the User	Bruno N. Costa "Bruno Nog"
review/helpfulness	Fraction of the users who found the review helpful	2/2
review/score	Rating given to the movie by the reviewer. Ranges from 0 to 5 stars	5

review/time	Time of the review in unix time. It can easily be converted to a date.	1209168000 (can be converted to Saturday, April 26, 2008 12:00:00 AM)
review/summary	Title/summary of the review	Certainly the most creative movie of the year
review/text	Actual text of the review	Amazing! The best musical film of this age! The Beatles are realy universal reference of good music, and now the songs in the movie show us a new understanding of them! Fantastic! I buy after have seen on the Movie and no regrets about the DVD. Good quality of images and sounds. Very interesting extras.

However, the dataset is missing some important information, such as the title of the movie being reviewed and its price (most reviews are about DVDs, as the dataset predates streaming). In order to solve this problem, we used metadata published by the authors. The metadata contains information about 48.19 million products (including all types of movies, not only movies). All the information we needed was readily available in the metadata (McCauley).

3.2 Data Formatting:

The data formatting of the dataset is highly unconventional. It does not follow standard types such as CSV or TSV. Instead, each review spans multiple lines, with an empty line separating two reviews. Each line has the following format:
variable/attributeName: variable/attribute value

We have one line for each attribute/variable, and as each review has 8 attributes/variables (the same 8 variables described in the subsection above). Therefore, each review spans around 8 lines. However, this formatting has many issues. First of all, there is plenty of redundancy. The name of each variable appears in each review. So, we will see each variable/attribute name around 8 million times.

Another important issue is that the dataset does not follow a standard character set, making it hard for the computer to read/decode and process the information. One cannot just use the standard methods to read files, but needs to incorporate error handling. There is also some inconsistency in our dataset, as not all reviews have all the variables/attributes. Last but not least, because of the size of the dataset, it is heavy, weighing almost 9 GB.

3.3 Data treatment:

In order to solve the problems described above, we treated the data to generate a dataset of a manageable size that follows a standard format. The format the group decided on was CSV (comma-separated values). In this format, the names of the attributes/variables are in the first line of the file. This way, we solve the redundancy problem described above, as variable/attribute names will appear only once. Moreover, each review spans a single line. In each line, the attributes/variables are separated by a comma character (,). The order of the names of the attributes in the first line is the same as the order

of the attributes in each other line; this way, the dataset follows a table-like structure. This format is easier to read and parse, and many programming languages have inbuilt methods to read CSVs.

In case the text of a review contains comma characters, we enclosed the entire review in quotation marks. This makes the CSV reader consider everything inside the quotation marks as a single attribute/variable. Last but not least, we trimmed the dataset, making it have a manageable number of reviews (as we have to process them all). We did that by only keeping reviews written in 2012 and only keeping movies with more than 50 reviews. In order to solve the problem we had with the character set of the review, we only kept reviews that follow a standard character set.

3.4 Data Cleaning:

Because of the issues of the dataset, we needed to be creative to clean it: we used Python dictionaries. We iterated through the dataset line by line. For each line we broke it into two, using the colon (:) as the delimiter. The part before the colon is the attribute/variable name, which was used as the key of the Python dictionary. The part after the colon was used as the value of the current key. Thus, following the format dictionary[variable/attributeName] = value. We also cleaned the data, first of all, we replaced all different types of space characters with a regular space character. We also replaced tab characters (\t) with a regular space character, removed new line characters (\n), and removed all single quotation mark (‘) and double quotation mark (“) characters (as they do interfere with the CSV). We saved each variable into a dictionary, and once we found the text of the review, we wrote the line with all the variables/attributes into the CSV file. This assumes that every review has review text, and that review text is the last field for each review. These are reasonable assumptions based on the nature of the dataset and the data description.

As for the metadata, we joined the cleaned dataset with the metadata based on the product ID. We read the metadata, which was in JSON into R, did a join based on product ID and included movie title, price, and average rating into the cleaned dataset.

3.5 New attributes:

We also derived the following attributes from our data and added them to our dataset:

- Average rating of movie (from 1 to 5)
- Word count of review
- Mean word count of reviews of a movie
- Median word count of reviews of a movie
- Sentiment Score for each individual review (value from 0 to 100, 0 being extremely negative and 100 being extremely positive)
- Sentiment Classification (Bad, Neutral, Good)
- Readability Score (Value from 0 to 100, 0 being hard to read)
- Standard deviation of score
- Standard deviation of sentiment score
- Rating number and review count (helped us clean the dataset)

4 Network Characteristics

To better understand the structural makeup of our movie network, we analyzed several key network characteristics. These metrics offer insights into how movies are interconnected through shared reviewer activity and help characterize the patterns within the network.

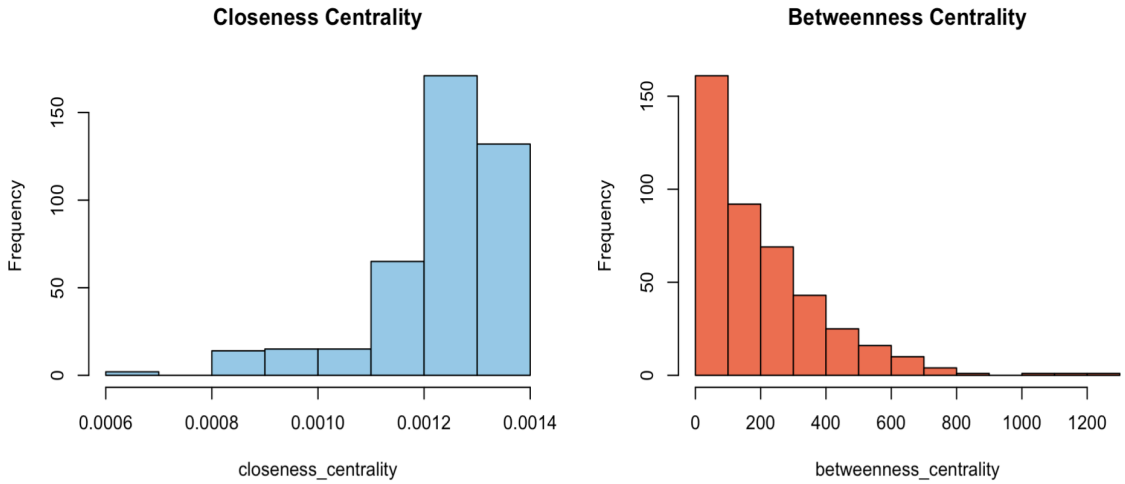
Table 2

Table with the basic network statistics of the network

Characteristics	**Values**
Density	0.23
Transitivity	0.59
Diameter	6
Mean Distance	1.99
Mean Degree	97.38

Figure 2

Histograms of the closeness and betweenness centrality of the network



The network density is 0.23, indicating that nearly one-quarter of all possible connections between movies are present (Table 1). This is a relatively high density for a projection of a bipartite network of 424 movies and suggests that the reviewer community is highly active, with many individuals reviewing multiple overlapping films. As a result, the network is densely knitted, with frequent co-reviewing forming the basis for strong inter-movie connectivity.

The transitivity, or global clustering coefficient, is 0.59, which is also a notably high value. This indicates that if a movie A is connected to both movie B and movie C, then B and C are also likely to be connected. Considering this in a practical context, it reflects strong local cohesion: clusters of movies that are commonly reviewed by the same groups of users tend to form tightly bound triangles. This clustering could suggest that reviewers gravitate toward certain themes, genres, or series, and potentially reflect thematic or interest-based communities within the network, which we explore more about later.

The diameter of the network is 6, meaning the greatest shortest path between any two movies is six steps. This measure captures the network's "reach", or how far apart the most distantly related movies are. A diameter of 6 is relatively small, particularly for a projected bipartite graph, and is consistent with small-world network characteristics. Each node may not necessarily have a large number of neighbors, but are connected to each other by a low number of edges. The mean distance of 1.99 further reinforces this: on average, any movie is just two steps away from any other. This reflects the strong interconnectivity of the movie ecosystem: shared reviewer activity acts as a bridge linking different movies. Together, these findings indicate that the movie network is tightly interwoven through shared

audiences, allowing rapid spread of information or influence through a relatively small number of connections.

The degree distribution provides additional insight into the network's structure. The mean degree is 97.38, indicating that, on average, each movie is connected to nearly 100 other movies through shared reviewers. This high average suggests strong reviewer overlap and again highlights the connected nature of the reviewer base. However, the distribution is right-skewed, meaning that while a core group of movies is highly connected, a larger number of movies have lower degrees, with over 150 movies connected to fewer than 50 others. This long tail reveals a core-focused structure in the network: a central hub of heavily reviewed titles surrounded by a broader collection of less widely reviewed movies.

To gain a deeper understanding of the role individual movies play in the broader network structure, we examined two key centrality measures: closeness centrality and betweenness centrality. These node-level metrics help reveal which movies are most influential in terms of accessibility and connectivity within the network.

Closeness centrality captures how quickly a node can reach all other nodes in the network via the shortest paths. In our context, it reflects how well-connected a movie is in terms of shared reviewer overlap with the rest of the network. Movies with high closeness centrality are only a few steps away from any other movie, meaning they share reviewers with many different parts of the network.

As shown in Figure 2 (left), we find that most movies exhibit relatively high closeness, suggesting that the reviewer base does not form isolated systems or silos of nodes. This could imply a broadly connected reviewer population, like one that creates cohesive bridges across different segments of the movie landscape.

Betweenness centrality, on the other hand, measures how often a node lies on the shortest path between other nodes. In other words, it identifies movies that serve as bridges connecting different parts of the network.

In our network, most movies do not serve this bridging function, exhibiting low betweenness centrality as shown in Figure 2 (right). This aligns with the overall high clustering and short average path length, as most movies are already closely connected, and the need for a bridging function is low. However, a small subset of movies demonstrates significantly higher betweenness, emerging as critical nodes that link otherwise loosely connected clusters. These may be genre-crossing or highly popular titles that appeal to multiple, distinct reviewer groups, or other types of films that serve this bridging function.

5 Sentiment & Readability Analysis

We considered two approaches to sentiment analysis, both of which are outlined in our paper.

5.1 Machine Learning Approach:

Our group considered using the BERT (Bidirectional Representation for Transformers) approach, which has some advantages over classic Sentiment Analysis methods. First of all, it does not read a word per se, it reads a word in context. Each word is within a context window (a number of words before and after the current word). The current word has its meaning modified by the words that come before it, and by the words after it in the context window (Hashemi-Pour., Lutkevich, 2024). So, for this approach, Carleton College is different from Carleton University, as the former is modified by the word College and the latter is modified by the word University.

This approach is more accurate than traditional approaches and can understand complex structures like sarcasm. What is more, there are free pretrained models online. Many were trained on IMDB data, which is good.

However, there are a few issues that prevented us from using BERT. First of all, it does not output a number saying how positive or negative the review is. It outputs a classification of the review instead. In most of the pretrained models, the classification is either only positive/negative or positive/neutral/negative. This makes our analysis lose some depth, as we are putting the reviews in fewer

boxes. What is more, neural networks are opaque; we just feed them a review, and they spit out a classification. However, what ultimately prevented us from using BERT was the fact that it can only handle short reviews, up to around 512 tokens (where each token is roughly a word). Our dataset contains many reviews that are thousands of words long. This led us to explore a second approach.

5.2 The dictionary approach:

In the dictionary approach, every word has a weight, which is either positive or negative. Positive weights are associated with a positive sentiment, and negative weights are associated with a negative sentiment. Some words have higher values than others, depending on how strong the sentiment associated with them is. With the dictionary approach, we read the text one word at a time, and sum the weight of the current word to the running sum, normalize it, and return it as a score from -1 (extremely negative) to 1 (extremely positive) (Eila, 2023). What is more, there are many tried and tested, high-quality dictionaries for this approach. Some dictionaries are general purpose, and some of them are fine-tuned for specific fields. Examples of dictionaries include Harvard-IV and Henry's Financial Dictionary (Proellochs, Feuerriegel, 2023). In order to implement this in R, we used the library Sentiment Analysis Vignette library (Proellochs, Feuerriegel, 2023). This library has functions that given text, it outputs a sentiment score (from -1 to 1). The library also has methods to convert this number into a classification (positive, negative, neutral).

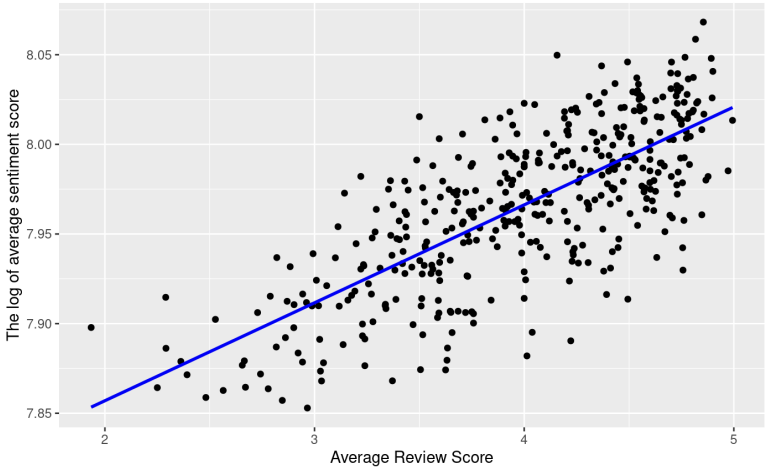
The main drawback of this approach is that words have a single meaning (their meaning is not modified by context). However, this approach has some key advantages over BERT. First of all, it can handle texts of an arbitrary length. What is more, it outputs a numeric score, which is easier to interpret than just a classification. Based on the sentiment score we know how positive the review is, and not just whether it is positive or not. Still, we can convert this score into a classification if we want. Based on our experiments, the dictionary approach was very accurate. However, for each review, to make the sentiment score closer to a percentage, we converted the sentiment score of that specific review, from -1 to 1 to a number between 0 to 100, where 0 is very negative, and 100 is very positive.

5.3 Statistical analysis based on sentiment analysis:

One of the first questions we tried to answer was whether the sentiment score (the numerical value from 0 to 100 assigned to each reaview) is related to the actual review score given to a movie by a user. Our question stems from the feeling that people are inclined to give extreme review scores (either 1 star or 5 stars). For example, imagine that someone gave 5 stars to a movie. However, while they liked the movie, they did not find it great/excellent. In this example, the fair score would be 3 stars. We want to see if the score expressed in the review is backed up by the sentiment of the text in the review. To answer this question, we fit a linear model on sentiment score and review score shown in Figure 3.

Figure 3

Plot of the linear regression model on the log of average sentiment score (per movie) vs. the average review score (per movie)



We found that there is a relationship between the log of the average sentiment score of a movie and the average review score of the movie. The coefficients of our model are $\beta_0 = 47.86300$, with standard error 0.25618, and $\beta_1 = 1.45878$, with standard error 0.06344. What is more, this is a representative model as $R^2 = 0.5562$ and the Adjusted $R^2 = 0.5551$. Therefore, over 55% of the variation of the log of the sentiment score of a movie can be explained by an increase in rating. With this model, a one-unit increase in the rating is associated with a 4.30071 time increase in the sentiment score.

Another question that we wanted to answer was whether reviews have a different length based on whether they are positive, negative, or neutral. We initially had the impression that positive reviews tended to be shorter. After all, they might just include a positive adjective or a compliment to the movie, while negative reviews might explain why the user thinks the movie is bad. For this, we considered reviews that gave the movie 1 or 2 stars negative, reviews that gave the movie 4 or 5 stars positive, and 3-star reviews as neutral. We did find a difference in the mean and the median number of words based on review rating, as shown in Table 3:

Table 3

Mean and median number of words for reviews considered good/bad/neutral based on the above criteria

Review Rating	Mean Word Count	Median Word Count
Good	114.8494	56
Bad	130.2309	70
Neutral	168.6968	92

We can see through Table 3 that positive reviews are indeed shorter, and negative reviews are longer in both mean and median. Surprisingly, the neutral reviews are the longest in mean and median. This came as a surprise. So, our hypothesis was only partially confirmed by this, as positive reviews are the shortest.

We wanted to do a similar analysis for the sentiment score. We wanted to know whether reviews with positive sentiment scores were shorter than reviews with negative sentiment scores. To do so, we used the sentiment classification provided by our sentiment analysis library: Sentiment Analysis Vignette (the sentiment classification is based on the sentiment score). Our results are shown in Table 4:

Table 4
Mean and median word count for movies classified as good/bad/neutral according to sentiment score

Review Sentiment Score	Mean Word Count	Median Word Count
Good	125.70594	63
Bad	131.16104	67
Neutral	70.13006	39

As we can see in table 4, the results are quite different when using sentiment scores instead of review scores. We still have that negative reviews are longer than positive reviews in both the mean and median. However, according to this measure, neutral reviews are the shortest. This might happen because the threshold for neutral reviews used by our sentiment analysis library (Sentiment Analysis Vignette) is much narrower than the threshold we used for ratings. We believe that many reviews that were considered neutral using review scores (reviews with a 3 star rating) are considered positive by the sentiment analysis library. This might be the reason the mean and median number of words of positive reviews using this measure are much higher. Still, we found that positive reviews are usually shorted than negative reviews.

5.4 Readability Score:

Another measure we are interested in is the readability score. The readability score is defined as how hard text is to read. The readability score is a number from 0 to 100, where 0 is extremely hard to read and 100 is extremely easy to read. The measure we used for this, known as Flesch-Kincaid (Eq 1). This measures readability based on sentence length and complexity. It uses number of words, number of sentences, number of syllables to define the complexity of a sentence (Readable, 2024).

$$206.835 - 1.015\left(\frac{\text{total words}}{\text{total sentences}}\right) - 84.6\left(\frac{\text{total syllables}}{\text{total words}}\right) \qquad \text{Eq. 1}$$

To implement this in our dataset, we used the library Quanteda, which, given text, automatically outputs the Flesch-Kincaid score (Benoit, Watanabe et al, 2018).

However, we did not find significant results using the readability score. For example, we tried to see if the sentiment classification of reviews (positive/negative/neutral) was related to the readability score. In other words, whether reviews were more readable if they were positive/negative/neutral. However, the results seemed normally distributed. We did not find any meaningful relationship between sentiment classification and readability score. An analogous analysis was done using actual user ratings, where 1 and 2 are considered negative scores, 4 and 5 are positive scores, and 3 is a neutral score. However, we also found no evidence that readability scores are related to review scores. Therefore, readability scores did not yield any meaningful results.

6 Exploratory Data Analysis

To complement our network-level analysis, we conducted exploratory data analysis (EDA) to examine how key movie-level covariates relate to degree centrality, a measure of how many direct reviewer-based connections a movie has in the network. This allows us to investigate how a movie's characteristics or reception might influence its structural role within our network.

We examined five main covariates: average rating, number of reviews, price, average review word count, and average sentiment score. Each plot and analysis aimed to uncover whether these

attributes are associated with a movie's network centrality, and by extension, its potential reach or visibility among shared reviewers.

Figure 4

Plots of the degree (per movie) vs 6 covariates: average rating, number of reviews, movie price, average word count, average sentiment score, and proportion of positive reviews (all per movie)

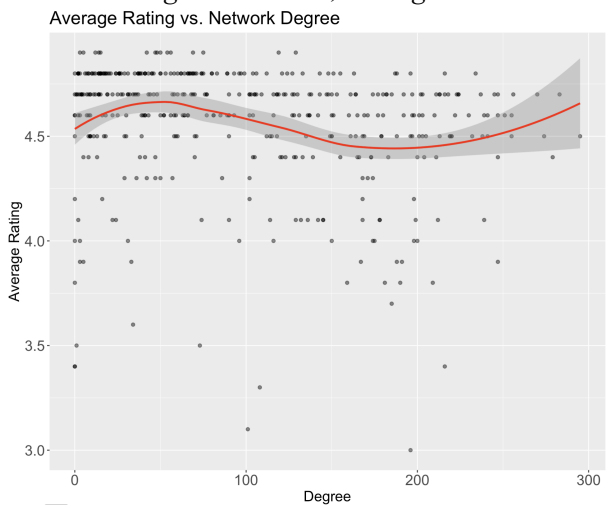


Figure 5

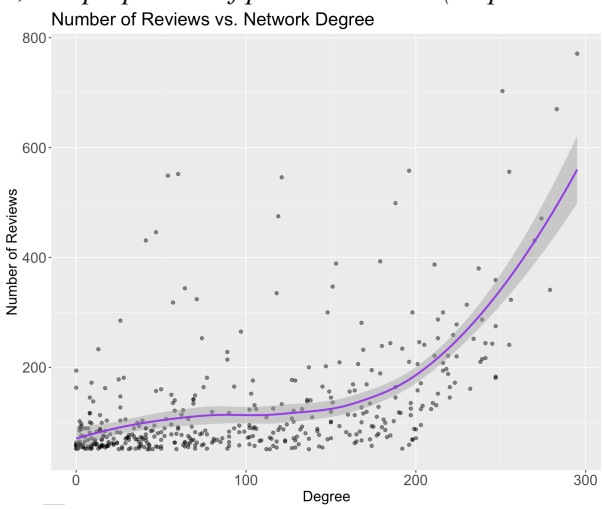


Figure 6

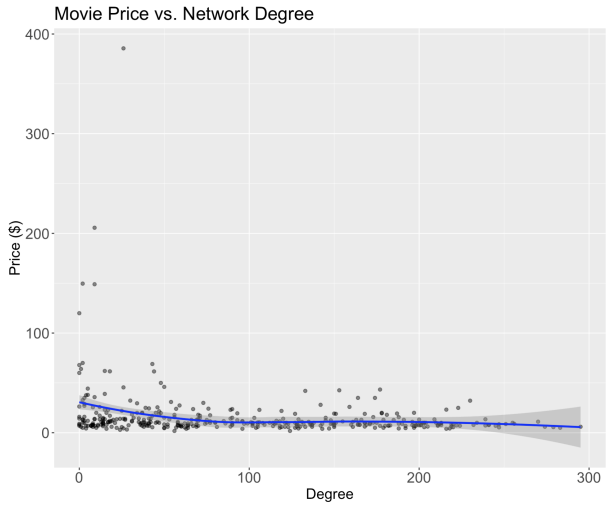


Figure 7

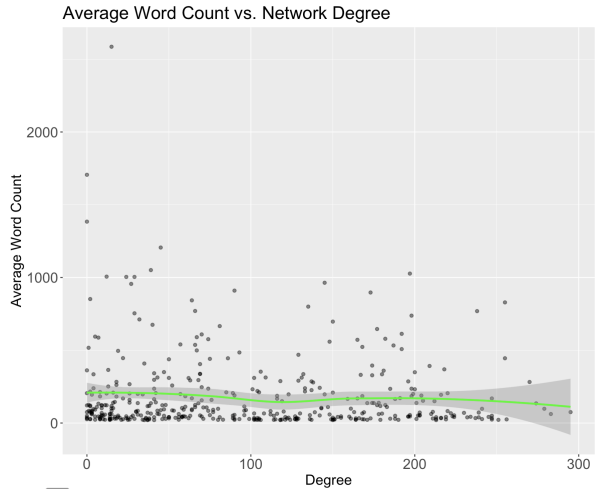


Figure 8

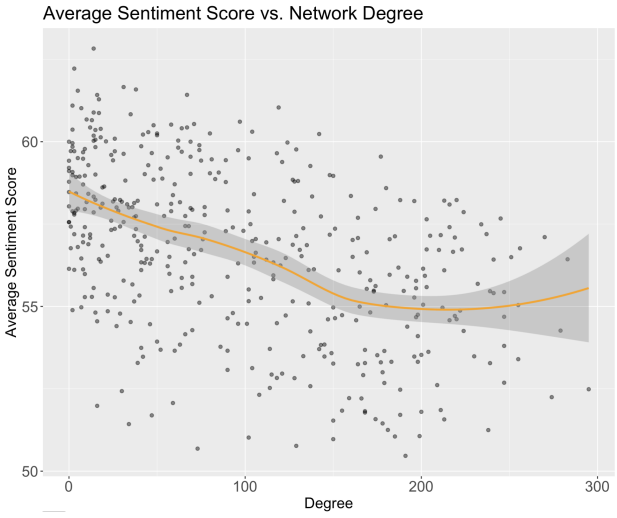
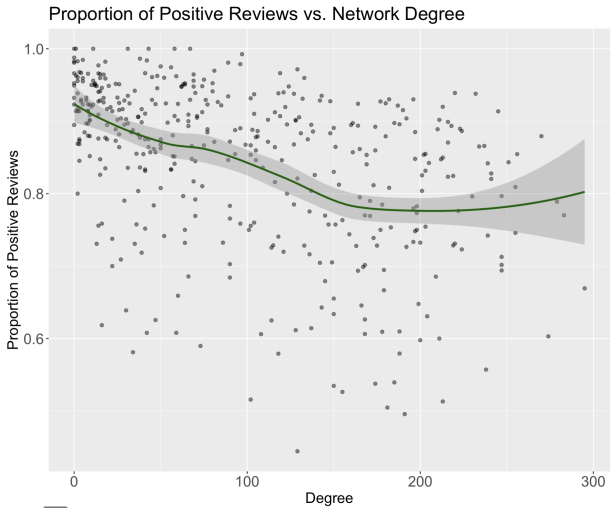


Figure 9



6.1 Rating vs. Degree

We began by analyzing whether higher-rated movies are more central in the network. The underlying assumption was that highly rated films might attract a wider audience, and thus form more reviewer-based connections with other movies. However, our plot in Figure 4 shows no strong relationship between average rating and degree centrality. This suggests that movie quality, as perceived by reviewers, does not directly correspond to greater connectivity. High ratings may reflect niche appeal or satisfaction within a smaller audience segment, rather than broader viewership overlap.

6.2 Review Count vs. Degree

Unsurprisingly, we observed a strong positive relationship between the number of reviews and degree centrality, as demonstrated in Figure 5. Movies that have been reviewed more frequently tend to have higher degrees, likely due to increased reviewer overlap with other films. This supports the idea that popular movies act as hubs in the network, playing a key role in linking different parts of the reviewer landscape.

6.3 Price vs. Degree

We explored whether accessibility, as approximated by price, influences a movie's centrality. One assumption was lower-priced movies, being more affordable, could attract a broader reviewer base and thus higher degree. However, our analysis found no clear trend between price and degree in Figure 6, and suggests that price does not serve as a barrier or gateway to centrality, at least within this dataset.

6.4 Word Count vs. Degree

Next, we examined whether movies that elicit longer reviews, potentially indicating more complex, emotional, or thought-provoking content, occupy different positions in the network. However, the relationship between average review word count and degree centrality appears negligible, as shown in Figure 7. This finding indicates that movies prompting more detailed or reflective responses are not necessarily more central, and reviewer verbosity does not translate to broader engagement across the network.

6.5 Sentiment Score vs. Degree

Finally, we assessed the relationship between average sentiment score (derived from review text) and degree. Somewhat counterintuitively, we observed a negative relationship, where movies with lower sentiment scores tend to have higher degrees shown in Figure 8. A similar downward trend was observed when using the proportion of positive reviews, as indicated in Figure 9. This could suggest that widely reviewed or more central movies receive more polarized or critical attention, perhaps because they attract a broader range of viewers with differing expectations.

7 Network Modeling

Exponential Random Graph Models (ERGMs), are a class of models derived from exponential family theory that can be applied to networks in order to gain probabilistic insights about the network and its attributes (Frank & Strauss, 1986). The general form of an ERGM, which we will describe as defined in *Exponential Random Graph Models (Ergms) Using Statnet*, is

$$P(Y = y) = (\frac{1}{k(\theta)})h(y)exp[\theta^{\top}g(y)], \qquad \text{Eq. 2}$$

where Y is the originally observed network, y is one realization of the network Y with the same number of vertices, $g(y)$ is a vertex of model statistics for the network of y, θ is the vector of coefficients for the model statistics, $h(y)$ defines the baseline behavior of the model when the values of the θ vector equal 0, and $k(\theta)$ is a normalization constant that allows for the probability to be between 0 and 1. The formula for $k(\theta)$, as stated in statnet's ERGM help file, is

$$k(\theta) = \sum_{y} h(y)exp[\theta^{\top}g(y)] \qquad \text{Eq. 3}$$

ERGMs are akin to logistic models in the way that they also model successes and failures. In equation 1, the originally observed network, Y, can be thought of as an adjacency matrix full of 1s and 0s that respectively correspond to the existent or non-existent edges in the network. The y term is then the same matrix as Y, but the 1s and 0s may be different. With a 1 being a success and a zero being a failure, we can think of an ERGM as many logistic models tied into one larger model. In fact, when we apply a log term to the numerator of the general form, notice that we get a linear formula:

$$log(exp[\theta^{\top}g(y)]) = \theta_1 g_1(y) + \theta_2 g_2(y) + \theta_3 g_3(y) + \cdots + \theta_p g_p(y), \qquad \text{Eq. 4}$$

where p is the number of predictors. This should provide some insight on the relationship between ERGMs and logistic models.

The general form of an ERGM can also be re-expressed to look at the conditional log-odds of an edge between two nodes.

$$logit\, P(Y_{ij} = 1 | y^{c}_{ij}) = \theta^{\top}\delta_{ij}(y) \qquad \text{Eq. 5}$$

Where Y_{ij} represents an edge connecting nodes i and j, y^{c}_{ij} represents the entire network excluding the edge of Y_{ij}, and $\delta_{ij}(y)$ is a vector of change statistics. Change statistics are the amount that $g(y)$ changes depending on whether the edge between nodes i and j is there or not, holding the rest of the network fixed. So, the log odds of the probability of nodes i and j being connected, given the state of the rest of the network, is equal to the coefficients times the change statistics for our covariates. Think of Y_{ij} as representing a single cell in the adjacency matrix for the network. The edge that this cell represents would be between nodes i and j, and we are interested in the single probability of this cell containing a 1 (or a success).

One key difference to keep in mind when working with ERGM models is that the network statistics are functions of the network (as implied by the notation of $g(y)$), meaning that they will have to be computed on the originally observed network. Therefore, it is important to apply a function,

represented by g, in the ERGM model to each statistic to let the model know how to calculate the statistic. This function can take many forms, usually separated into three variations:

1. **Main Order** statistics are measures that deal with the general increase or decrease of a value across nodes. Given nodes i and j, and estimate θ, we can calculate these statistics with $\theta * (x_i + x_j)$, for any quantitative covariate x. For example, between nodes i and j, a total higher movie price might be related to that movie being more or less likely to form an edge with another movie depending on whether θ is negative or positive.
2. **Second Order** statistics are measures that deal with similarities or differences between two node values. We can calculate these statistics with $\theta * |x_i - x_j|$, for any quantitative covariate x. For example, a smaller difference between nodes i and j in movie price may have a smaller or larger probability of being connected depending on whether θ is negative or positive.
3. **Structural** statistics measure network characteristics such as transitivity, centrality, or clustering. One example is the edges term which is also known as the "intercept" term for ERGMs. It ensures that the model includes the same number of edges as the original network.

It is important to note that for categorical covariates, these statistics will act differently than aforementioned. Since our model only considers quantitative covariates, we do not need to consider any categorical statistics or interpretations.

7.1 Valued ERGMs

As our network includes weighted edges, it is important to consider valued ERGMs as a potential tool to explain our network. The main difference between a valued ERGM and a non-valued ERGM is that the valued ERGM is supported over many valued graphs, and valued graphs have a larger support set. In the non-valued ERGMs previously discussed, we are concerned with either the presence of an edge (1) or the absence of an edge (0), making the ERGM binary. For a valued ERGM, we are concerned with the value of an edge ranging from an absence (0) to some larger value (i.e. 30). Therefore, we need to consider how the values are distributed. Hence, we will need to reference a similar distribution in our model. The basic form for a valued ERGM is

$$P_{h,g}(Y = y; \theta) = \left(\frac{1}{k(\theta)_{h,g}}\right) h(y) exp[\theta^{\top} g(y)],\ y \in S, \qquad \text{Eq. 6}$$

where S is the support and $k(\theta)_{h,g}$ is the normalizing constant (Krivitsky 2012). Since we are no longer concerned with binary terms, we must calculate new terms with different methods. For example, rather than using the edges term as one would use for the intercept of a non-valued ERGM, one would use the sum term, which ensures that the total edge count for the modeled network is the same as the original network.

Using the *ergm.count* library in R, we tested multiple valued ERGMs (Krivitsky 2024). In order to build a valued ERGM model, a response is needed for the edge values, and a reference distribution is needed to approximate the distribution of these edge values. The weighted edges of how many users reviewed both movies were used as the response, and a geometric distribution was found to be the best fit for our reference distribution. That being said, processing the valued ERGMs would sometimes take days to run, and when our network attributes were added to the model, the model did not run in about 2 days. The interpretation of the model estimates are also more complicated for valued ERGMs. For these two reasons, we decided to stick with the non-valued ERGM as it was faster to compute and could still reveal interesting trends in our network.

7.2 Building a Model

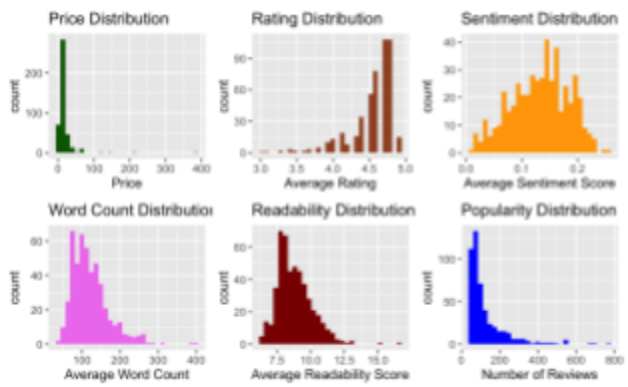
The ERGM modeling process was initiated by creating a random intercept model and then slowly adding variables to the model and checking to see how the AIC changed. Covariates considered for the model included main and second order terms for the price, average rating, word count, popularity,

sentiment score, and readability predictors. For structural predictors, we included the edges term, which is very important to include as it acts to maintain the same amount of edges in the simulated network as the original network, and therefore maintaining the same density. We also considered the Geometrically Weighted Edgewise Shared Partners (GWESP) term which is a weighted measure of transitivity. The GWESP term requires a decay value to be specified. A smaller decay value influences the model to encourage tighter clustering while a large decay value promotes less clustering.

Transformations were applied to variables based on their distributions derived from the histograms found in Figure 10. Ultimately, log transformations were applied to the readability, price, word count, and popularity predictors as all of them had right-skewed distributions. Sentiment score and average rating were left-skewed, and therefore, not transformed.

Figure 10

A series of histograms showing the distributions of our covariates



The first issue encountered when building the model was that there was missing data for the price variable. ERGM models are unable to handle NAs, and therefore they must be removed or replaced with filler data. About 10% of the movies had missing data on their price. There were multiple ways that we could handle this issue. One was to remove the movies with missing data from the model and continue. The other was to impute the average price of a movie. And finally, we could decide not to include the price variable in the final model. We decided to remove the movies with missing data and continue our modeling process for multiple reasons. The first reason is that we are interested in how the price of a movie affects the odds of a tie, and therefore, would like to include the price covariate in our model. Secondly, once we removed the movies with missing data from the model, we still had about 380 movies remaining as opposed to 424. We determined that 380 movies is a substantial enough amount of data to get conclusive results from.

After removing movies with missing data and transforming our variables, we tested a model with main and second order effects for all of our covariates, as well as the edges and GWESP structural terms. We found that the only insignificant predictor was the second order term for readability score. After removing this term from the model, our AIC did not change, but the BIC decreased justifying its removal. We also found that a smaller decay value for the GWESP term resulted in a lower AIC value.

Using a decay value of 0.01 for the GWESP term resulted in our best model in terms of AIC, which is what we decided on as our final model:

$$log\left(\frac{P(Y_{ij}=1|y^c_{ij})}{P(Y_{ij}=0|y^c_{ij})}\right) = \theta_1 \cdot edges + \theta_2 \cdot [log(read_i) + log(read_j)] + \theta_3 \cdot (sentiment_i + sentiment_j) +$$
$$\theta_4 \cdot |sentiment_i - sentiment_j| + \theta_5 \cdot [log(price_i) + log(price_j)] + \theta_6 \cdot |log(price_i) - log(price_j)| + \qquad \text{Eq. 7}$$
$$\theta_7 \cdot (avg_rating_i + avg_rating_j) + \theta_8 \cdot |avg_rating_i - avg_rating_j| + \theta_9 \cdot [log(word_i) + log(word_j)] +$$
$$\theta_{10} \cdot |log(word_i) - log(word_j)| + \theta_{11} \cdot [log(count_i) + log(count_j)] + \theta_{12} \cdot |log(count_i) - log(count_j)| + \theta_{13} \cdot GWESP_{0.01}$$

7.3 The Final Model

Fitted ERGMs communicate a lot of useful information about relationships within the network. Parameter estimates can be interpreted similarly to a logistic regression model so that a one unit change in the predictor results in a change in the log odds. In this case the log odds will refer to whether there exists a tie between two nodes (1) or not (0). As a result of the model building process, a final model was decided on. The AIC value was 58897. The coefficients for each term and interpretations for them can be seen in Table 5.

Table 5
Model output for each term as well as their interpretations

Term	Type	Estimate	P-value	Interpretation
Edges	Structural	-7.47	<0.0001	When all other terms have values of zero, the probability of the observed network occurring is 0.06%.
GWESP	Structural	-0.92	<0.0001	As the number of shared partners between two movies increases, the probability of the movies being connected decreases*.
log(Price)	Main order	-0.32	<0.0001	A two time increase in the total price between two movies is associated with an average decrease of 20% in the odds of a user reviewing both movies*.
log(Price)	Second order	-0.25	<0.0001	A two time increase in the difference in prices between two movies is associated with an average decrease of 16% in the odds of a user reviewing both movies*.
log(Word Count)	Main order	1.03	<0.0001	A two time increase in the average word count of reviews between two movies is associated with an average increase of 104% in the odds of a user reviewing both movies*.
log(Word Count)	Second order	-0.20	<0.0001	A two time increase in the difference in average review word count between two movies is associated with an average decrease of 13% in the odds of a user reviewing both movies*.
log(Popularity)	Main order	1.02	<0.0001	A two time increase in the number of reviews between two movies is associated with an average increase of 103% in the odds of a user reviewing both movies*.
log(Popularity)	Second order	-0.26	<0.0001	A two time increase in the difference in the number of reviews between two movies is associated with an average decrease of 16% in the odds of a user reviewing both movies*.
Rating	Main order	-0.49	<0.0001	A total one score increase in the rating of two movies is associated with an average decrease of 39% in the odds of a user reviewing both movies*.
Rating	Second order	-0.74	<0.0001	A decrease of 1 unit in the difference between two movies is associated with an average increase of 52% in the odds of a user reviewing both movies*.

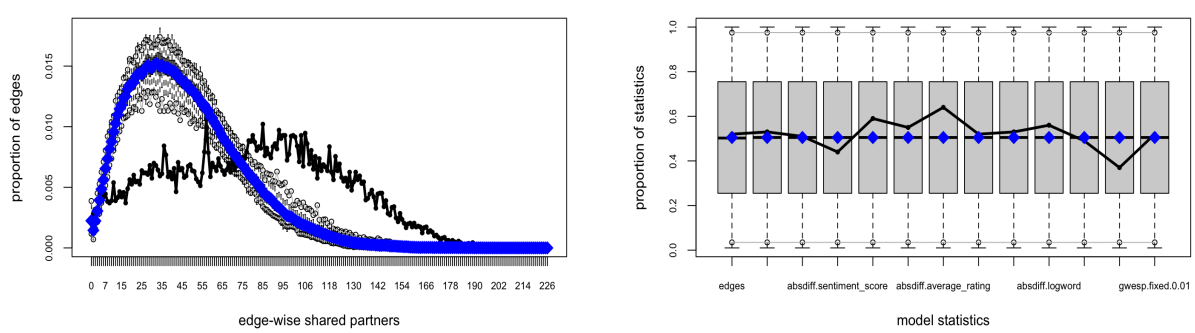
Sentiment Score	Main order	-6.53	<0.0001	A total 0.1 unit increase in the sentiment score of two movies is associated with an average decrease of 48% in the odds of a user reviewing both movies*.
Sentiment Score	Second order	-3.12	<0.0001	A difference of 0.1 in sentiment score between two movies is associated with an average decrease of 27% in the odds of a user reviewing both movies*.
log(Readability Score)	Main order	-0.87	<0.0001	A two time increase in the total readability score between two movies is associated with an average decrease of 45% in the odds of a user reviewing both movies*.

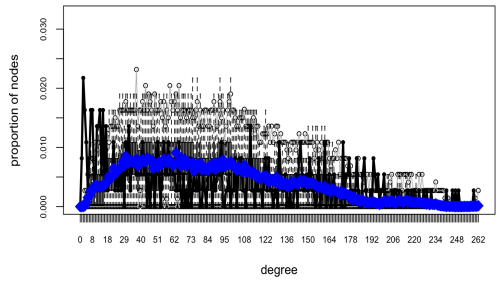
*Holding all else constant

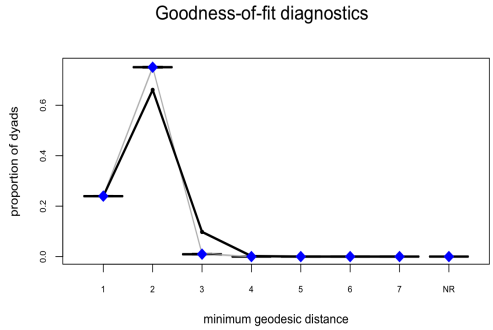
7.4 Goodness of Fit

A low AIC value is not the only thing of interest when developing a model. After developing ERGMs, it is also very important to check the goodness of fit as a way of assessing our model. The current method of goodness of fit tests is to simulate many networks from the developed model and then compare the characteristics of these simulations to the characteristics of the originally observed network (Kolaczyk & Csárdi 2020). The final model decided upon has the GOF results shown in Figure 11. The black line represents the values from our originally observed network, while the boxplots represent the distribution of values from our simulated networks. The blue point represents the median. From the figure, we can see that our model actually performs pretty well when looking at the model statistics and minimum geodesic distance plots. The degree plot shows some dissimilarities between our observed network and the simulated ones, but the general trend is well emulated by the model. In terms of edge-wise shared partners, our model performs very poorly. We see that the model over predicted the proportion of edges that have less shared partners and under predicted the proportion of edges that have more shared partners. In order to solve this, we attempted to add in structural predictors to the model that could potentially explain some of this transitivity and clustering pattern. Some terms tested were k-star, GWDegree, and GWDSP terms. Unfortunately, any addition of these terms to the model resulted in infeasible computation times. Therefore, we were unable to improve the fit of our model.

Figure 11
Goodness of Fit Plots







Although our model may not be the best predictive tool, it provides helpful information that gives light to some trends in our network. The covariate interpretations may be especially helpful in giving light to what generally affects whether movies are reviewed by an identical user or not.

8 Community Detection

In social network analysis, community detection refers to the process of identifying subgroups of nodes that are more densely connected to each other than to the rest of the network (Luke, 2015). These subgroups can reveal higher-order patterns that are not visible at the level of individual nodes or dyads, making them useful for uncovering latent structures in complex networks (Luke, 2015).

A key concept underlying many community detection methods is modularity, a statistic that measures the extent to which the observed density of edges within groups exceeds what would be expected if connections were formed at random (Newman, 2006; Luke, 2015). A high modularity score suggests that the network has a strong community structure, where nodes within the same group are more interconnected than expected by chance.

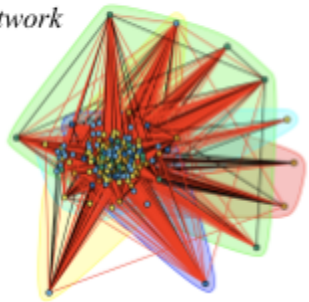
Building on these principles, we applied community detection to the movie network to investigate its broader structural properties. We aimed to understand whether groups of movies with shared audiences cluster together in meaningful ways, and whether these clusters reflected known attributes of the movies.

8.1 Louvain Algorithm and Attribute Analysis

We applied the Louvain algorithm to the projected movie network using the *igraph* package in R (Csardi & Nepusz, 2006; Csárdi et al., 2025). The Louvain algorithm is a modularity optimization algorithm widely used for detecting communities in large-scale networks due to its scalability and efficiency (Blondel et al., 2008). It optimizes modularity by iteratively assigning nodes to communities by locally optimizing modularity, then collapses those communities into super-nodes and repeats the process on the reduced graph until modularity no longer improves (Blondel et al., 2008).

Figure 12

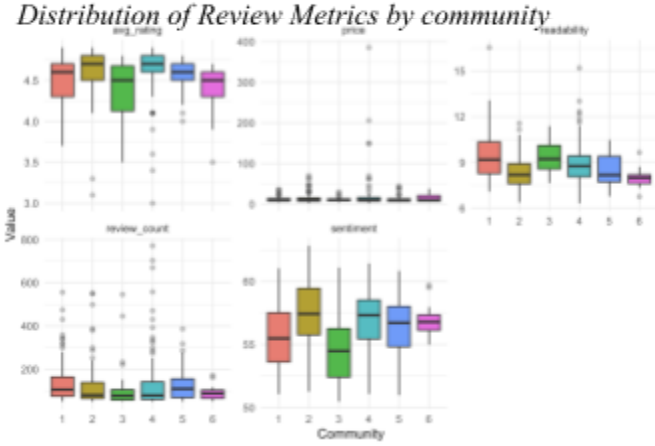
Louvain Communities in Movie Network



Once the communities were identified as shown in Figure 12, we examined whether certain movie-level attributes might explain their formation. However, our exploratory analysis in Figure 13 showed no consistent or interpretable patterns across communities with respect to these variables. This absence of clear trends suggested that community formation might not be explained by these attributes alone, prompting a further evaluation of the significance of the observed community structure.

Figure 13

Distribution of Review Metrics by community



8.2 Evaluating Modularity Against Random Network Models

To assess whether the community structure observed in the movie network was meaningful or could have emerged by chance, we compared the modularity of our empirical network to that of randomly generated networks using *igraph* (Csardi & Nepusz, 2006; Csárdi et al., 2025). As a baseline, we generated 500 Erdős–Rényi random graphs with the same number of nodes and edges as the original network (Newman, 2010), where edges are formed entirely at random.

For a more structurally realistic comparison, we generated 500 small-world networks using the Watts–Strogatz model (Watts and Strogatz, 1998). This model starts with a regular ring lattice and rewires edges with probability p, introducing shortcuts that reduce average path length while preserving local clustering (Luke, 2015). We calibrated the rewiring probability to match the empirical network's average path length and clustering coefficient, finding that $p = 0.06$ provided the best alignment.

We then applied the Louvain algorithm to each synthetic network and recorded their modularity scores as shown in Figure 14. The empirical movie network consistently exhibited higher modularity than all Erdős–Rényi networks, strongly suggesting that the observed community structure is not a product of random connectivity. However, its modularity was lower than that of small-world networks, implying that

while the network displays meaningful community organization, it does not conform neatly to the structure of typical small-world systems. Instead, it may reflect a more intricate form of organization, influenced by patterns not fully captured by conventional models of network formation.

Figure 14

Comparison of Original Network Modularity with Modularity Distributions from Random and Small-World Networks

Note. Left: Modularity scores from 500 Erdős–Rényi networks. Right: Modularity scores from 500 Watts–Strogatz small-world networks. The original network's modularity is overlaid in both.

9 Conclusion

In analyzing Amazon movie reviews from 2012, we have come across some interesting relationships. The original bipartite network between reviewers and movies was projected so that each movie represented a node and was connected by an edge when they were reviewed by a common user. Edge counts represented the number of common reviewers between two movies. With a relatively high density of 0.23, we realized that our network was much more clustered than predicted. This tells us that there may be more users reviewing multiple movies in the same time period than originally thought. Although our data subset of 424 movies only represented a small fraction of popular movies on Amazon, we still found there to be a lot of clustering. We found 10 isolated movies that were not connected to any other movies, but the rest of the movies were all included in one large, and very dense, cluster.

As might be expected, we found that more popular movies had a higher degree, meaning they had more connections with other movies. This makes sense as the number of reviews is related to the number of reviewers which is the potential number of people who could also review another movie, hence creating an edge. Interestingly, we also found that a decrease in average sentiment score was related to a higher degree, but it is difficult to interpret the significance of this relationship.

After developing our ERGM model, we found the general results that

- Increases in readability score, sentiment score, rating, and price were associated with a decrease in the odds of an edge.
- Increases in popularity and word count were associated with an increase in the odds of an edge forming.
- Decreases in the difference between values of sentiment score, popularity, word count, rating, and price were associated with an increase in the odds of an edge forming.

A main takeaway from this is that movies with more similar values, in terms of our covariates, are more likely to be reviewed by a common user.

Community detection found that we can find some trends in movies of similar genres being clustered together. For example, one community detected contained mostly fitness DVDs.

With more computational power, we could do a more comprehensive analysis of Amazon movie reviews as we would be able to analyze a larger network that contains more data, run ERGM models with more terms to improve our model's fit, and use more accurate sentiment and readability text analysis calculations.

We hope that this analysis allows for a better understanding of what goes into someone deciding to review two different movies, contributes to methodological approaches for analyzing large-scale bipartite networks and provides a foundation for further research in recommender systems, marketing, and behavioral analysis.

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